Assessment Tool - 3

Descriptive Written Test.

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1. Intelligent Agents in a Smart Home Automation System.

INTRODUCTION:

* A Smart home automation system uses [AI] to control lighting, temperature and security based on user behavior and environmental conditions. It receives real-time inputs from sensors such as motion detectors and temperature sensors. Intelligent agents help the system make decisions automatically, improve comfort, save energy, and maintain security.

TYPES OF INTELLIGENT AGENTS:

1. Simple Reflex Agent : Acts based on current inputs using rules.

* Switches on lights when motion is detected.

* Activates an alarm when an intruder is detected.

2. Model-Based Agent: Uses current inputs and past information.
   * Remembers room occupancy.
   * Tracks temperature and door activity.

3. Goal-Based Agent: Takes actions to achieve specific goals.
   * Maintains comfortable temperature.
   * Reduces electricity consumption.

4. Utility-Based Agent: Chooses the best action by balancing different needs.
   * Balances comfort and energy savings.
   * Maintains security without disturbing occupants.

BEST APPROACH:

* A hybrid Approach combining model-based, goal-based, and utility-

based agents is most effective. It considers user behaviour, achieves goals, and balances comfort, energy efficiency and security.

CONCLUSION:

* A hybrid intelligent agent makes the smart home more comfortable, energy-efficient, secure, and responsive to user needs.

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2. UNIFORMED SEARCH STRATEGIES FOR A ROBOT IN AN UNKNOWN MAZE:

INTRODUCTION:

* A Robot placed in an unknown maze must find a path from its starting position to the exit without knowing the maze layout in advance. AI uses uninformed search strategies to explore possible paths systematically. Two important strategies are uniform cost search [UCS] and iterative Deepening Search [IDS].

1. UNIFORM COST SEARCH (UCS).

* UCS explores the path with the lowest total cost first.

Working:

* Starts from the initial position.
* Explores the lowerest path
* Continues until it reaches the exit.

Advantages: Finds the least-cost path when step costs are non-negative.

Disadvantages: Requires more memory and may take more time.

Time and Space: Can be exponential in the solution cost.

2. ITERATIVE DEEPENING SEARCH [IDS]:

* IDS repeatedly searches with increasing depth limits until it finds the exit.

working:

* Starts with depth limit 0.
* Increases the limit step by step.
* Stops when the exit is found.

Advantages: Uses less memory and finds shallow solutions.

Disadvantages: Repeats some searches and does not guarantee the least-cost path when step costs differ.

Time complexity: $O(b^d)$

Space complexity: $O(bd)$

* Here, b is the branching factor and d is the depth of the solution.

Relation to Intelligent Agents:

- SIMPLE REFLEX AGENT: Reacts to nearby walls and obstacles.
- MODEL-BASED AGENT: Remembers explored paths and obstacles.

- GOAL - BASED AGENT: Plans actions to reach the exit.
- UTILITY - BASED AGENT: Chooses a route based on cost, time, and energy.

BEST APPROACH:

* A Model - based and goal - based agent combined with UCS is effective when paths have different costs. The model - based agent remembers explored areas, while UCS finds the least - cost route.

* If memory is limited and all movements have equal costs, IDS is a suitable alternative.

CONCLUSION:

* UCS is best for finding the least - cost path, while IDS is suitable for memory - limited environments. Combining intelligent agents with an appropriate search strategy helps the robot reach the exist efficiently.